

Aim : To develop Software requirement Specification (SRS) in IEEE format for project

Objective : To prepare a SRS for Hotel management system.

Following is the Detail explained SRS (Software Requirement Specification) for our Project that Hotel is Management system.

HOTEL MANAGEMENT SYSTEM

Software Requirement Specification

- **Introduction**

The Hotel Management System is a tool for booking the rooms of Hotel through online by the Customer. It provides the proper management tools and easy access to the customer information.

- **Purpose**

This Hotel Management System Software Requirement Specification (SRS) main objective is to provide a base for the foundation of the project. It gives a comprehensive view of how the system is supposed to work and what is to be expected by the end users. Client's expectation and requirements are analyzed to produce specific unambiguous functional and non-functional requirements, so they can be used by development team with clear understanding to build a system as per end user needs.

This SRS for HMS can also be used for future as basis for detailed understanding on how project was started. It provides a blueprint to upcoming new developers and maintenance teams to assist in maintaining and modifying this project as per required changeability.

- **Scope of the Project**

The HMS project is intended for the reservations for room that can be made through online. It will be able to automate the various operations of the Hotel. Our Hotel Management System will have three end users: Customer, Receptionist, and Hotel Manager. Hotel Management System will consist of Booking Management System, DBMS Server, and Report Generator. Customers will

be able to check for room's availability, select the rooms, and pay for the room. Receptionist will have access to update or modify booking details. Manager will be able to view the financial report and be able to update room information such as cost and category.

The main goal of this introduced automated HMS software is to simplify every day process of hotel. Day to day Hotels are increasing and they need to automate to provide customer ease of access. It will be able to take care of services to customer in a quick manner. This automation will be able to replace the drawbacks of large customer information physical files which were difficult to handle. Secure Transaction, quick retrieval of information, ease of use, quick recovery of errors, fault tolerance are some of the benefits that development team will be working on to achieve end user satisfaction.

- **Definitions, Acronyms and abbreviations**

SRS	Software Requirement Specifications
HMS	Hotel Management System
DBMS	Database Management System
FR	Functional Requirement
NFR	Non Functional Requirement

- **References**

- Software Engineering 9th Edition, Ian Sommerville
- Fundamentals of Database System, 6th Edition, Ramez Elmasri, Shamkant B. Navathe

- **Overview**

The remaining sections of this documentations describes the overall descriptions which includes product perspective and functions, characteristics of users. It also consists of Assumptions, and Constraints. Overall description is listed in section 2. Section 3 includes Specific Requirements which consists of Functional and Non-functional requirements, External Interface Requirements, Software System Attributes, Performance Requirements, Capacity Requirements, Availability Requirements, Safety Requirements and Requirement Traceability Matrix.

OVERALL DESCRIPTION

- **Product Perspective**

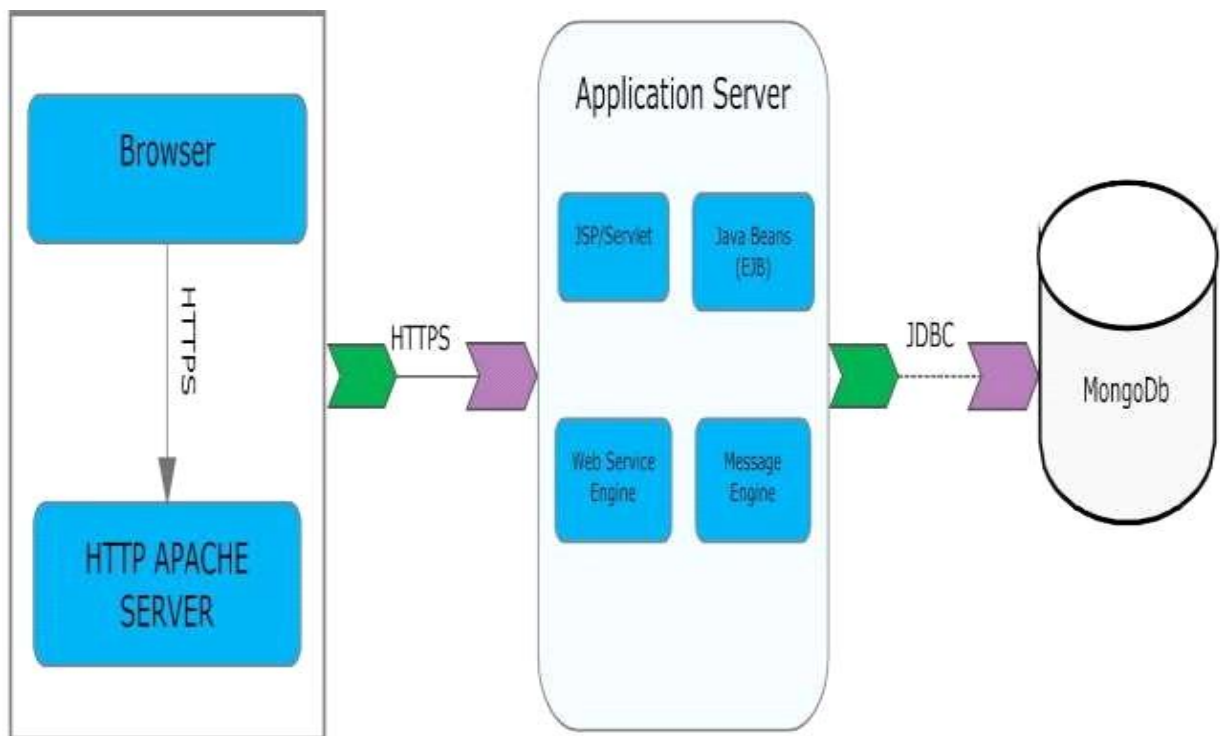


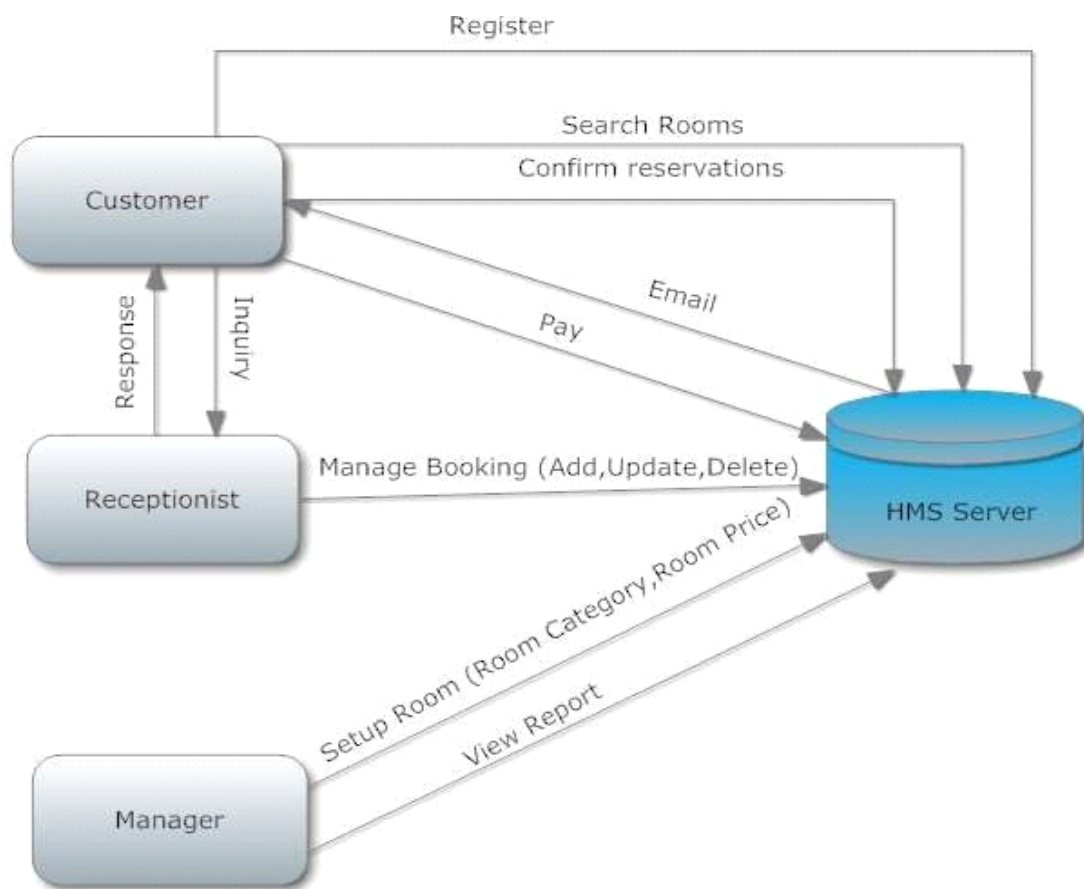
Figure 2.1 Product Perspective

- **Product Functions**

Our Product General functions are:

- Customer Registration
- Check for Availability Of Rooms
- Display the Rate

- Confirmation Of Booking
- Email Notification
- Payment
- Set Room Details
- Manage Booking Details
- Generate Report
- Customer Service



*Figure 2.2
System
Architecture*

- **User Characteristics**

There are 3 user Levels in our Hotel Management System:

- Hotel Manager
- Receptionist
- Customers

Hotel Manager

Manager have every access to the hotel system. Manager is solely responsible for managing hotelresources and staffs. Manager can view any report such as financial report, customer information,booking information, and room information, analyze them and take the decision accordingly. Manger is required to have experience on managing hotel previously, and have base knowledge ofdatabase and application server.

Receptionist

Hotel Receptionist sole purpose is to provide the quality customer service. She have least access than manager. She can manage the booking details. She can search for availability of rooms, add the customer, confirm the booking, and update the booking details. Manager of hotel would probably want the receptionist who have good communication skills and command over English language. She should have basic IT Knowledge.

Customer

Customer are vital part of the system. Customer have access to view the vacant room information and price range. They should be able to confirm the booking and cancel it if necessary. Customers have access to customer service desk portal to forward their inquiry. Customer should at least be capable to use the web UI interface.

- **Constraints**

- **Memory:** System will have only 10GB space of data server.
- **Language Requirement:** Software must be only in English.
- **Budget Constraint:** Due to limited budget, HMS is intended to very simple and just for basic functionalities. UI is going to be very simple.
- **Implementation Constraint:** Application should be based on Java only.
- **Reliability Requirements:** System should sync frequently to backup server in order to avoid the data loss during failure, so it can be recovered.

- **Assumption and Dependencies**

It is assumed that system developed will work perfectly that's going to be developed under the Windows OS, and Apache Server with Mongo DB database. If in case of any difficulties, SRS should be flexible enough to change accordingly.

- **SPECIFIC REQUIREMENTS**

- **External Interface Requirements**

- **User Interfaces**

The user interface for system shall be compatible to any type of web browser such as MozillaFirefox, Google Chrome, and Internet Explorer.

- **Hardware Interfaces**

Server Side			
Monitor	Processor	RAM	Disk Space
Resolution: 1024x768	Intel or AMD 2GHZ	4GB	10GB
Client Side			
Monitor	Processor	RAM	Disk Space
Resolution: 1024x768	Intel or AMD 1GHZ	512MB	2GB

- **Communication Interfaces**

The System shall be using HTTP/HTTPS for communication over Internet and for intranet communications, it shall use TCP/IP protocol.

- **Functional Requirements**

- **Registration**

FR1. The Customer should be able to register with their details

FR2. The system should record following customer details into member database

FR3. The system shall send verification message to email

- **Logging In**

FR4. The system should verify the customer email & password against the member database when logging in

FR5. After login, member should be directed to Home screen

- **Reservation**

FR6. The system should enable customer to check for availability of rooms

FR7. The system should display rate for all rooms

FR8. The system should allow customer to confirm or cancel the booking

FR9. The system should record booking details into database

- **Receptionist Access**

FR10. The system should allow Receptionist to update, add or delete booking information

FR11. The system should provide customer desk portal access to receptionist for providing response to customer inquiry

- **Manager Access**

FR12. The system should generate financial and customer report for manager

FR13. The system should enable manager full modification access to customer ,booking and room information

- **Payment Management System**

FR14. The system should allow customer to pay bill via online using credit or debit card

- **Performance Requirements**

NF1. Data in database should be updated within 2

seconds. **NF2.** Query results must return results

within 5 seconds **NF3.** Load time of UI Should

not take more than 2 seconds **NF4.** Login

Validation should be done within 3 seconds

NF5. Response to customer inquiry must be done within 5 minutes.

- **Security Requirements**

NF6. All external communications between the data's server and client must be encrypted

NF7. All data must be stored, protected or protectively marked.

NF8. Payment Process should use HTTP over Secure protocol to secure the payment transactions

- **Safety Requirements**

NF9. Database should be backed up every hour.

NF10. Under failure, system should be able to come back at normal operation under an hour.

- **Capacity Requirements**

NF11. Not more than 10,000 members to be registered

NF12. System need to handle at least 20 transactions during peak hours.

- **Availability Requirements**

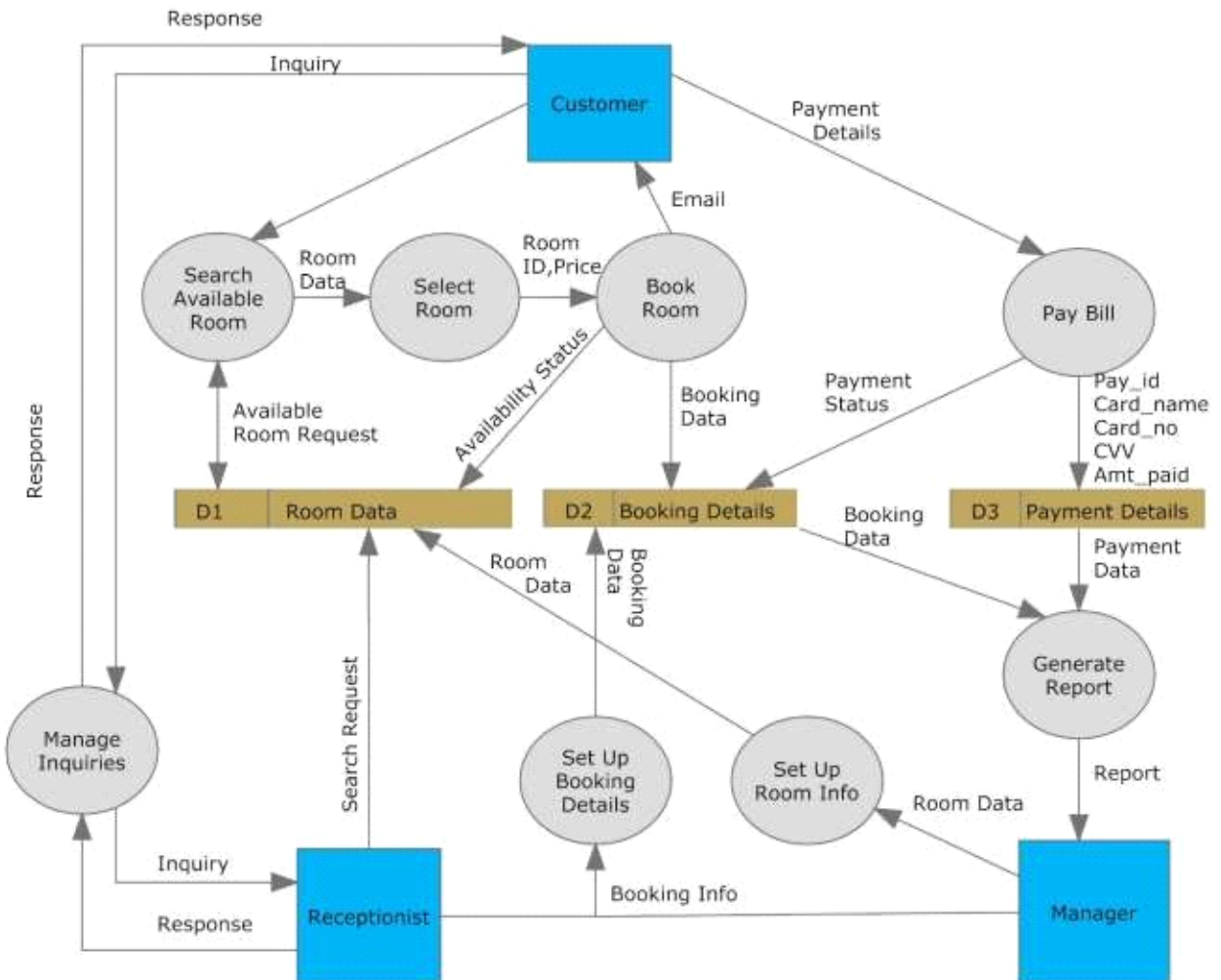
NF13. Report should be generated automatically every day for manager and anytime upon request.

- **Software System Attributes**

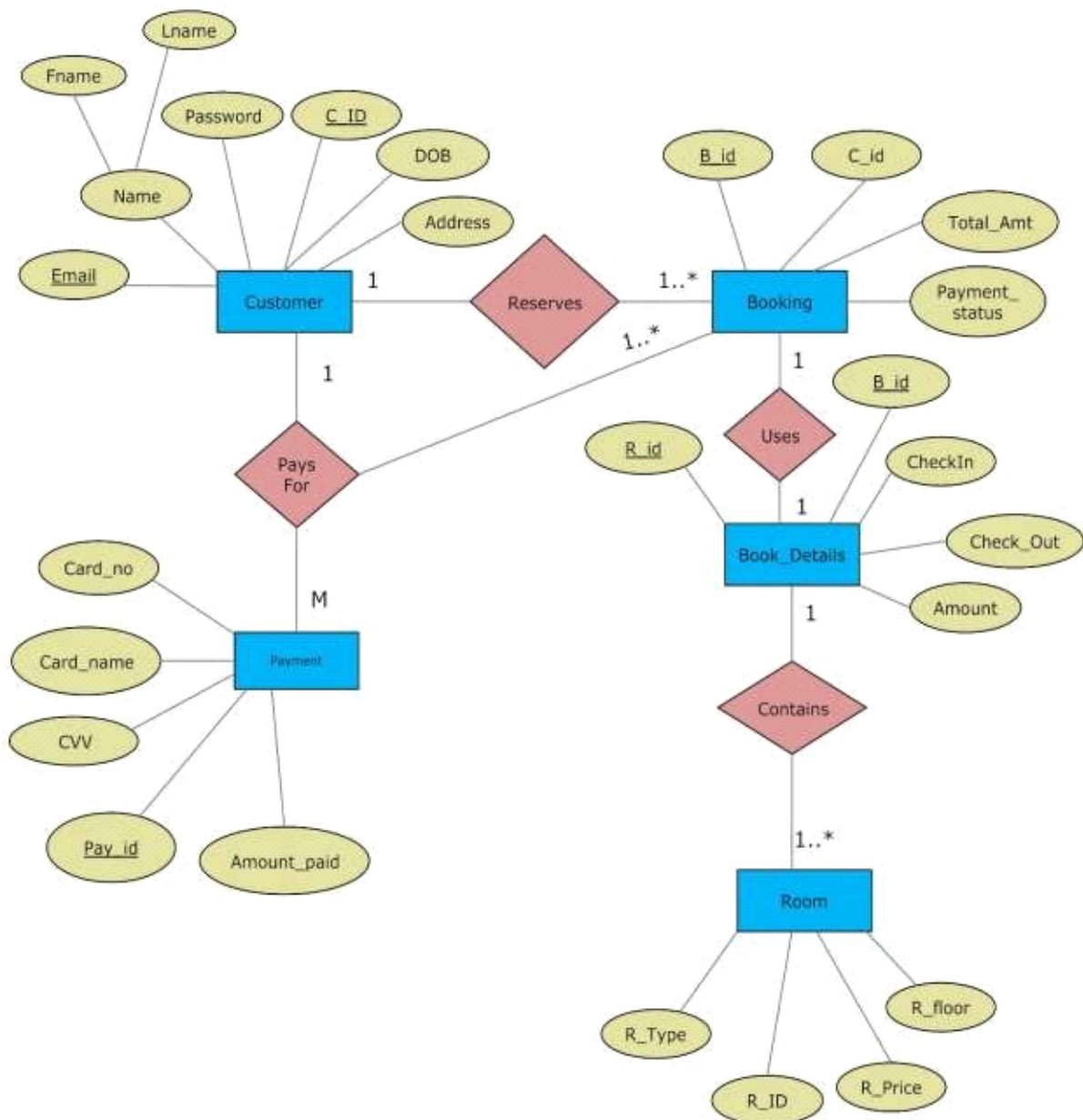
- **Correctness:** This system should satisfy the normal regular Hotel Management operations precisely to fulfill the end user objectives
- **Efficiency:** Enough resources to be implemented to achieve the particular task efficiently without any hassle.
- **Flexibility:** System should be flexible enough to provide space to add new features and to handle them conveniently
- **Integrity:** System should focus on securing the customer information and avoid data losses as much as possible
- **Portability:** The system should run in any Microsoft windows environment.
- **Usability:** The system should provide user manual to every level of users.
- **Testability:** The system should be able to be tested to confirm the performance and client specifications.
- **Maintainability:** The system should be maintainable.

APPENDIXES

DATA FLOW DIAGRAM



ENTITY RELATIONSHIP DIAGRAM



Conclusion :

We have successfully prepared SRS document for Currency Detector for the Visually Impaired.